

ICDFA MODULE 8: STEGANOGRAPHY AND HIDDEN DATA ANALYSIS PRACTICAL REPORT

Student Details

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1. INTRODUCTION

Steganography is the practice of concealing information within another digital object, known as a **carrier file**, in such a way that the existence of the hidden information may not be immediately obvious. Unlike encryption, which transforms information into an unreadable form but normally makes it apparent that protected data exists, steganography attempts to conceal the existence of the information itself.

In this practical, I acted as a junior digital forensic trainee conducting a controlled review of an image suspected of containing hidden information. The laboratory activities were performed only on an authorised carrier image and files created within the training environment.

The practical involved preparing and preserving an original carrier image, creating controlled evidence files, embedding hidden data using Steghide, extracting concealed information using the correct password, verifying integrity through cryptographic hashes, comparing the original and modified images, and performing a controlled password dictionary recovery test using StegSeek.

The practical also included an independent exercise in which I created my own hidden evidence file, selected a password, embedded the file, extracted it, verified its integrity, and tested the password-recovery workflow using a controlled wordlist.

2. OBJECTIVES

The objectives of this practical were to:

- Explain the difference between steganography and encryption.
- Describe insertion and substitution-based hiding techniques.
- Explain the Least Significant Bit (LSB) concept.
- Prepare and preserve an authorised carrier image.
- Create a controlled secret evidence file.
- Calculate SHA-256 hashes for evidence preservation and integrity verification.
- Use Steghide to embed a secret file inside an image.
- Extract concealed information using the correct password.

- Compare original and extracted evidence files.
- Compare the original carrier image with the resulting stego image.
- Perform a controlled password dictionary test using StegSeek.
- Create and test an independently generated hidden evidence file.
- Document all practical activities using screenshots and command evidence.

3. PRE-LAB REVIEW AND LABORATORY PREPARATION

Before beginning the practical, I reviewed the Module 8 lessons and study material on **Steganography and Hidden Data Analysis**.

The topics reviewed included:

- Hidden information and data concealment.
- Steganography.
- Encryption.
- Insertion-based hiding techniques.
- Substitution-based hiding techniques.
- Least Significant Bits (LSB).
- Bitmap images.
- Steghide.
- Data extraction.
- Password protection.
- Controlled password recovery.

The practical environment was prepared using Linux laboratory system.

A dedicated working directory was created to ensure that the original carrier image, secret evidence files, extracted files, hashes, and wordlists remained organised and separated.

The following directory structure was used:

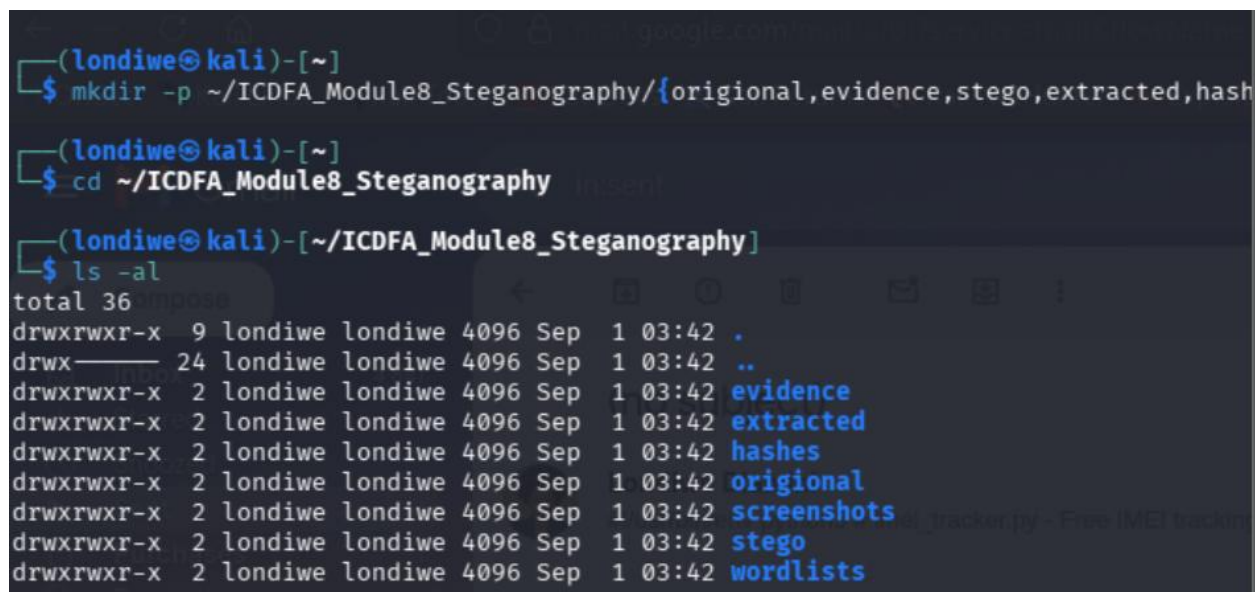
```
mkdir -p
~/ICDFA_Module8_Steganography/{original,evidence,stego,extracted,hashes,wordlists,screenshots}
```

```
cd ~/ICDFA_Module8_Steganography
```

The directories were organised as follows:

- **original** – preserved original carrier image.
- **evidence** – controlled evidence files created during the practical.
- **stego** – images containing embedded information.
- **extracted** – files recovered from the stego image.
- **hashes** – SHA-256 hash records.
- **wordlists** – controlled password wordlists.
- **screenshots** – screenshots collected as evidence of the practical.

SCREENSHOT 1: Module 8 study material/pre-lab review and laboratory folder structure



```
(londiwe@kali)-[~]
$ mkdir -p ~/ICDFA_Module8_Steganography/{original,evidence,stego,extracted,hashes,wordlists,screenshots}

(londiwe@kali)-[~]
$ cd ~/ICDFA_Module8_Steganography

(londiwe@kali)-[~/ICDFA_Module8_Steganography]
$ ls -al
total 36
drwxrwxr-x  9 londiwe londiwe 4096 Sep  1 03:42 .
drwx----- 24 londiwe londiwe 4096 Sep  1 03:42 ..
drwxrwxr-x  2 londiwe londiwe 4096 Sep  1 03:42 evidence
drwxrwxr-x  2 londiwe londiwe 4096 Sep  1 03:42 extracted
drwxrwxr-x  2 londiwe londiwe 4096 Sep  1 03:42 hashes
drwxrwxr-x  2 londiwe londiwe 4096 Sep  1 03:42 original
drwxrwxr-x  2 londiwe londiwe 4096 Sep  1 03:42 screenshots
drwxrwxr-x  2 londiwe londiwe 4096 Sep  1 03:42 stego
drwxrwxr-x  2 londiwe londiwe 4096 Sep  1 03:42 wordlists
```

4. THEORETICAL BACKGROUND

4.1 Steganography

Steganography is the process of hiding information inside another file or medium. The file that carries the hidden information is known as the **carrier file** or **cover object**.

For example, a text file may be hidden inside an image. To an ordinary user, the image may appear to be a normal picture. However, hidden data may exist within its underlying digital structure.

The resulting file containing both the carrier information and the concealed data is commonly referred to as a **stego object** or **stego image**.

The main objective of steganography is to reduce suspicion by concealing the existence of the information.

4.2 Difference Between Steganography and Encryption

Steganography and encryption are both techniques used to protect information, but they operate differently.

Encryption transforms readable information into an unreadable format. An observer may know that encrypted data exists but cannot easily understand the information without the appropriate decryption key or password.

Steganography, on the other hand, attempts to conceal the existence of the information itself by hiding it inside another object.

In this practical, Steghide was used to conceal a controlled evidence file within an authorised image. The hidden information was also protected using a password.

Therefore, steganography and encryption can complement one another. Steganography hides the presence of information, while password protection or encryption helps prevent unauthorised recovery of the concealed data.

4.3 Insertion-Based Hiding

Insertion-based hiding involves adding hidden information to an existing digital object.

Depending on the technique used, information may be placed into areas of a file that do not significantly affect the normal appearance or operation of the carrier.

From a forensic perspective, insertion-based hiding is important because an apparently normal file may contain additional information that is not immediately visible during ordinary use.

4.4 Substitution-Based Hiding

Substitution-based hiding involves replacing selected portions of the original digital data with information representing the secret message.

A common example is modification of individual bits within digital media.

The changes may be small enough that the human eye or ear does not easily detect them.

4.5 Least Significant Bit (LSB)

Digital information is represented using binary values consisting of bits.

The **Least Significant Bit (LSB)** is the bit with the lowest value within a binary representation.

For example, changing the least significant bit of a value may produce only a very small numerical difference. In digital images, carefully modifying selected least significant bits may result in changes that are visually difficult for a human observer to detect.

This concept is important in steganography because small changes to image data can potentially be used to represent hidden information while preserving a visually similar appearance.

From a forensic perspective, this means that two images may appear almost identical to a human observer while having different underlying binary data and different cryptographic hashes.

5. PART A – PREPARE AND PRESERVE THE EVIDENCE

5.1 Objective

The purpose of this section was to obtain the authorised laboratory carrier image, preserve the original copy, confirm its file type, and calculate a SHA-256 hash before any steganographic activity was performed.

The original image was preserved and was not overwritten during the embedding exercise.

5.2 Procedure

The authorised BMP carrier image was downloaded from the laboratory source and saved inside the `original` directory.

The image was referred to in this report as:

`tower_original_image_for_lab.bmp`

If the downloaded filename was different, the actual filename used in the laboratory should be substituted throughout the report.

The file was first copied or saved into the original evidence directory.

The file type was then examined using:

```
cd ~/ICDFA_Module8_Steganography/original  
file tower_original_image_for_lab.bmp
```

The output was reviewed to confirm that the file was recognised as a BMP image.

The SHA-256 hash of the original carrier image was then calculated and recorded:

```
sha256sum tower_original_image_for_lab.bmp |  
tee ../hashes/original_carrier_sha256.txt
```

The recorded hash was preserved as the integrity reference for the original carrier image.

[INSERT SCREENSHOT 2: Original carrier image, file type confirmation and SHA-256 hash]

5.3 Forensic Interpretation

Preserving the original carrier image is important because the original evidence should remain unchanged.

The original image provides a baseline for later comparison with the stego image.

The SHA-256 hash acts as an integrity reference. If the original file is accidentally modified, a recalculated hash would normally differ from the original recorded value.

Forensic examination should therefore be performed on copies or newly created working files where possible, while the original evidence is preserved.

6. PART B – CREATE THE SECRET EVIDENCE FILE

6.1 Objective

The purpose of this task was to create a controlled text file named `secret.txt` for use as the hidden evidence file.

The file was created specifically for the ICDFA laboratory.

6.2 Procedure

The evidence directory was opened:

```
cd ~/ICDFA_Module8_Steganography/evidence
```

The secret evidence file was created using a text editor:

```
nano secret.txt
```

The following information was included in the file:

Full Name: [INSERT YOUR FULL NAME]

Controlled Evidence Message:

This file contains controlled evidence created for steganography testing in an authorised laboratory environment.

Lab Confirmation:

This file was created for the ICDFA Module 8 Steganography and Hidden Data Analysis practical.

The file was saved and its contents were displayed:

```
cat secret.txt
```

A SHA-256 hash was then calculated:

```
sha256sum secret.txt | tee ../hashes/secret_original_sha256.txt
```

The hash value was recorded for later comparison with the extracted file.

[INSERT SCREENSHOT 3: secret.txt contents and SHA-256 hash]

6.3 Forensic Interpretation

The SHA-256 hash generated at this stage represents the integrity value of the original secret evidence file.

After the file is embedded and extracted, the recovered file should be compared against this original hash.

Matching SHA-256 values provide evidence that the extracted file has the same content as the original controlled evidence file.

7. PART C – EMBED THE HIDDEN FILE WITH STEGHIDE

7.1 Objective

The purpose of this task was to embed `secret.txt` into a new stego image while preserving the original carrier image.

The laboratory password used for the guided exercise was:

1234

This password was specifically required for the controlled dictionary recovery exercise later in the practical.

7.2 Procedure

A new stego image was created rather than modifying the original carrier image.

The following command structure was used:

```
cd ~/ICDFA_Module8_Steganography

steghide embed ¥
-ef evidence/secret.txt ¥
-cf original/tower_original_image_for_lab.bmp ¥
-sf stego/tower_stego_lab4.bmp ¥
-p 1234
```

The command embedded the secret evidence file inside the carrier image and created the new file:

```
tower_stego_lab4.bmp
```

The original carrier image remained unchanged in the original directory.

The resulting stego image was then examined:

```
file stego/tower_stego_lab4.bmp
ls -lh original/tower_original_image_for_lab.bmp
stego/tower_stego_lab4.bmp
```

The SHA-256 hash of the stego image was calculated:

```
sha256sum stego/tower_stego_lab4.bmp | tee
hashes/stego_image_sha256.txt
```

7.3 Forensic Interpretation

The stego image may look visually similar to the original carrier image because steganographic techniques are designed to conceal information without making obvious changes to the carrier.

However, the resulting stego image contains modified digital information.

For this reason, the SHA-256 hash of the stego image is expected to differ from the SHA-256 hash of the original carrier image.

A different hash does not necessarily indicate corruption. In this case, the difference is expected because the stego image was intentionally modified through the embedding process.

8. PART D – EXTRACT AND VERIFY THE HIDDEN EVIDENCE

8.1 Objective

The purpose of this task was to recover the concealed evidence file using the correct password and verify that the recovered file matched the original.

8.2 Procedure

An extraction directory was created:

```
mkdir -p ~/ICDFA_Module8_Steganography/extracted
```

The hidden file was extracted using Steghide and the correct password:

```
cd ~/ICDFA_Module8_Steganography
```

```
steghide extract ¥  
-sf stego/tower_stego_lab4.bmp ¥  
-xf extracted/extracted_secret.txt ¥
```

```
-p 1234
```

The contents of the extracted file were displayed:

```
cat extracted/extracted_secret.txt
```

The original and extracted files were compared using:

```
diff evidence/secret.txt extracted/extracted_secret.txt
```

The SHA-256 hash of the extracted file was calculated:

```
sha256sum extracted/extracted_secret.txt | tee  
hashes/extracted_secret_sha256.txt
```

The original and extracted hash records were then displayed:

```
cat hashes/secret_original_sha256.txt  
cat hashes/extracted_secret_sha256.txt
```

8.3 Forensic Interpretation

The successful extraction demonstrated that the correct password allowed the concealed file to be recovered from the stego image.

The `diff` comparison was used to identify differences between the original and extracted files.

If no differences were reported and the SHA-256 hashes matched, this indicated that the extracted evidence file was identical to the original controlled evidence file.

The integrity verification process is important in digital forensics because recovered evidence should be examined in a way that allows the investigator to demonstrate whether its contents have remained unchanged.

9. PART E – COMPARE THE ORIGINAL AND STEGO IMAGES

9.1 Objective

The purpose of this task was to compare the original carrier image and the stego image.

The comparison focused on:

- File size.
- SHA-256 hash.
- Initial bytes.
- Readable strings where applicable.

9.2 Compare File Sizes

The sizes of both images were displayed:

```
ls -lh ¥  
original/tower_original_image_for_lab.bmp ¥  
stego/tower_stego_lab4.bmp
```

The file sizes were recorded and compared.

9.3 Compare SHA-256 Hashes

The SHA-256 hashes of both files were calculated:

```
sha256sum ¥  
original/tower_original_image_for_lab.bmp ¥  
stego/tower_stego_lab4.bmp
```

9.4 Compare Initial Bytes

The initial bytes of the original carrier image were examined:

```
xxd -l 64 original/tower_original_image_for_lab.bmp
```

The same number of bytes was then examined from the stego image:

```
xxd -l 64 stego/tower_stego_lab4.bmp
```

[INSERT SCREENSHOT 10: xxd comparison of the original and stego image]

9.5 Examine Readable Strings

Readable strings were examined where applicable:

```
strings original/tower_original_image_for_lab.bmp | head -n 20
```

The same process was performed on the stego image:

```
strings stego/tower_stego_lab4.bmp | head -n 20
```

[INSERT SCREENSHOT 11: strings comparison of the original and stego images]

9.6 Forensic Interpretation

The original carrier image and the stego image may appear visually similar because the purpose of steganography is to conceal information while minimising obvious changes to the carrier.

However, their underlying digital contents may be different.

The SHA-256 hashes are expected to differ because embedding information modifies the resulting digital file.

The file sizes may also differ depending on the steganographic method and the amount of hidden information.

Examining the initial bytes with xxd can provide a basic view of the binary structure at the beginning of each file.

The strings command may reveal readable text stored within the files, although the absence of visible strings does not prove that no hidden information exists.

These comparisons demonstrate an important forensic principle: **visual similarity does not mean digital identity**.

10. PART F – CONTROLLED PASSWORD DICTIONARY TEST

10.1 Objective

The purpose of this task was to demonstrate a controlled password-recovery workflow using a small laboratory wordlist.

The wordlist contained the known laboratory password:

1234

The test was performed only against the authorised stego image created during the practical.

10.2 Create the Laboratory Wordlist

A dedicated wordlist directory was used:

```
cd ~/ICDFA_Module8_Steganography/wordlists
```

A small controlled wordlist was created:

```
nano lab4_wordlist.txt
```

Example contents:

```
password  
123  
1234  
admin  
lab  
secret
```

The file was saved and displayed:

```
cat lab4_wordlist.txt
```

10.3 Run StegSeek

The controlled dictionary test was performed against the authorised stego image:

```
cd ~/ICDFA_Module8_Steganography
```

```
stegseek stego/tower_stego_lab4.bmp wordlists/lab4_wordlist.txt
```

The result was documented.

[INSERT SCREENSHOT 13: StegSeek result showing the password-recovery workflow]

10.4 Tool Failure Procedure

If StegSeek was unavailable or failed in the laboratory environment, the error was documented clearly.

For example:

```
stegseek --version
```

If the tool was not installed or could not run, the resulting error message was captured.

The Steghide embedding and extraction workflow was still completed successfully using the correct laboratory password.

10.5 Forensic Interpretation

The controlled dictionary test demonstrated how a known password can be included in a small wordlist and tested against an authorised stego image.

This exercise was limited to laboratory-created evidence and did not involve unauthorised access to external files.

From a forensic perspective, password-recovery workflows may assist an investigator in examining protected or concealed information when the investigation is properly authorised.

The quality and relevance of the wordlist can influence whether a password is successfully identified.

11. PART G – INDEPENDENT STUDENT TASK

11.1 Objective

The purpose of this section was to independently apply the skills learned during the guided practical.

A new hidden evidence file was created using the required naming format:

```
firstname_lastname_hidden_note.txt
```

11.2 Create the Independent Hidden Evidence File

The file was created inside the evidence directory:

```
cd ~/ICDFA_Module8_Steganography/evidence  
nano firstname_lastname_hidden_note.txt
```

The following explanation was included:

What is Steganography and Why Does It Matter in Digital Forensics?

Steganography is the practice of concealing information inside another digital object so that the presence of the hidden information may not be immediately obvious. A file such as an image can act as a carrier for hidden text or other data.

Steganography matters in digital forensics because a file that appears ordinary may contain concealed evidence. An investigator may therefore need to examine file structure, metadata, hashes, binary content, and possible hidden-data techniques when analysing suspicious digital media.

The ability to identify, extract and verify hidden information is important because concealed data may be relevant to an authorised investigation. At the same time, forensic analysis must preserve evidence integrity and operate within the authorised scope of the investigation.

The file was saved and its hash was calculated:

```
sha256sum firstname_lastname_hidden_note.txt ¥  
| tee ../hashes/independent_note_original_sha256.txt
```

11.3 Embed the Independent Evidence File

A password of my own choice was selected for the independent exercise.

Selected Password:

The file was embedded into a new stego image:

```
cd ~/ICDFA_Module8_Steganography
```

```
steghide embed ¥  
-ef evidence/firstname_lastname_hidden_note.txt ¥  
-cf original/tower_original_image_for_lab.bmp ¥  
-sf stego/firstname_lastname_stego.bmp ¥
```

```
-p YOUR_CHOSEN_PASSWORD
```

11.4 Extract the Independent Evidence File

The file was extracted using the selected password:

```
steghide extract ¥  
-sf stego/firstname_lastname_stego.bmp ¥  
-xf extracted/firstname_lastname_extracted_note.txt ¥  
-p YOUR_CHOSEN_PASSWORD
```

The extracted file was displayed:

```
cat extracted/firstname_lastname_extracted_note.txt
```

11.5 Verify Integrity

The SHA-256 hash of the extracted file was calculated:

```
sha256sum extracted/firstname_lastname_extracted_note.txt ¥  
| tee hashes/independent_note_extracted_sha256.txt
```

The original and extracted hashes were compared:

```
cat hashes/independent_note_original_sha256.txt  
cat hashes/independent_note_extracted_sha256.txt
```

The files were also compared using:

```
diff ¥  
evidence/firstname_lastname_hidden_note.txt ¥
```

extracted/firstname_lastname_extracted_note.txt

11.6 Create a Custom Wordlist

A controlled wordlist containing the selected password was created:

```
cd ~/ICDFA_Module8_Steganography/wordlists  
nano firstname_lastname_wordlist.txt
```

The selected password was included together with a small number of controlled test entries.

Example:

```
test  
password  
lab  
YOUR_CHOSEN_PASSWORD  
forensics
```

11.7 Test the Recovery Workflow

The controlled password-recovery workflow was tested against the independently created stego image:

```
cd ~/ICDFA_Module8_Steganography  
  
stegseek ¥  
stego/firstname_lastname_stego.bmp ¥  
wordlists/firstname_lastname_wordlist.txt
```

The result was documented.

12. EVIDENCE INTEGRITY TABLE

The following table should be completed using the actual SHA-256 values obtained during the practical.

Evidence Item	Original SHA-256	Extracted/Result SHA-256	Result
secret.txt			
extracted_secret.txt			
firstname_lastname_hidden_note.txt			
firstname_lastname_extracted_note.txt			
Original carrier image			
Guided stego image			
Independent stego image			

13. FINDINGS

The practical demonstrated that hidden information can be embedded inside an authorised carrier image while the resulting image may remain visually similar to the original.

The original carrier image was preserved and hashed before the embedding process. A controlled evidence file was then created and embedded into a newly created stego image using Steghide.

The hidden file was successfully extracted using the correct password.

The original and extracted evidence files were compared using diff and SHA-256 hashes. Matching results confirmed that the recovered evidence matched the original controlled file.

The comparison between the original carrier image and the stego image demonstrated that visually similar files can have different underlying digital content. This was demonstrated through differences in cryptographic hashes and binary comparison.

A controlled password dictionary workflow was also tested using a wordlist containing the known laboratory password.

The independent exercise demonstrated the ability to create, embed, extract and verify a separate hidden evidence file using a password selected by the student.

14. CHALLENGES ENCOUNTERED

Potential challenges encountered during the practical included:

- Ensuring that the original carrier image was not overwritten.
- Using the correct file paths.
- Ensuring that Steghide was installed and available.
- Remembering the correct password during extraction.
- Comparing hash values accurately.
- Ensuring that the controlled wordlist contained the required password.
- Tool availability or compatibility issues with StegSeek.

Where StegSeek was unavailable or produced an error, the error was documented as evidence. The successful Steghide embedding and extraction process remained the core practical requirement.

Any actual error encountered during the practical should be described below:

Actual challenge/error encountered: I could not locate the image file to use for the practical part of the assignment.

How the issue was addressed: Reached out to my class rep and other colleagues, they did not have assistance at the time.

15. FORENSIC INTERPRETATION

This practical demonstrated that digital evidence may contain information that is not visible through normal use of a file.

An image may appear visually normal while containing concealed data or modified binary information. For this reason, forensic analysis should not rely only on visual examination.

Cryptographic hashing provides an important method for documenting file integrity. In this practical, hashes were calculated before and after extraction to verify whether the recovered evidence matched the original controlled evidence file.

The preservation of the original carrier image was also important. The original image acted as the baseline evidence item, while separate stego images were created for embedding activities.

The use of controlled wordlists demonstrated a repeatable password-recovery workflow within an authorised laboratory environment.

Overall, the practical reinforced the importance of:

- Preserving original evidence.
- Working on controlled copies.
- Documenting commands and results.
- Recording cryptographic hashes.
- Comparing original and recovered files.
- Maintaining a repeatable forensic process.
- Operating only within an authorised scope.

16. CONCLUSION

This practical provided practical experience in the concepts of steganography and hidden-data analysis.

The activities demonstrated the difference between steganography and encryption and introduced concepts such as insertion-based hiding, substitution-based hiding and Least Significant Bit techniques.

The authorised carrier image was preserved and hashed before modification. Controlled evidence files were created and embedded into newly generated stego images using Steghide.

The hidden files were successfully extracted using the correct passwords and their integrity was verified through SHA-256 hashing and file comparison.

The original carrier image and stego image were also compared using file sizes, cryptographic hashes, hexadecimal output and readable strings.

The controlled StegSeek exercise demonstrated a password dictionary recovery workflow using a laboratory-created wordlist.

Finally, the independent task demonstrated the ability to apply the complete workflow independently by creating, embedding, extracting, verifying and testing a new hidden evidence file.

The practical demonstrated that visually similar files may contain different underlying digital information and reinforced the importance of systematic evidence preservation, integrity verification and authorised forensic examination.

FINAL FORENSIC SAFETY STATEMENT

All activities described in this report were performed only on the authorised laboratory carrier image and files created within the training environment. The password dictionary

tests were performed only against stego images created for this practical. No unauthorised files, systems or third-party data were accessed or tested.